Eye Gesture-Based Robotic Arm Control Using EOG and Machine Learning

Mohammed Alshehhi
Electrical Eng. and Computing Dept.
Rochester Institute of Technology
Dubai, UAE
maa5829@g.rit.edu

Abhilasha Singh
Electrical Eng. and Computing Dept.
Rochester Institute of Technology
Dubai, UAE
axscad5@rit.edu

Jinane Mounsef
Electrical Eng. and Computing Dept.
Rochester Institute of Technology
Dubai, UAE
jmbcad@rit.edu

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***Abstract*—This study introduces a novel machine learning-based framework for real-time, hands-free robotic arm control using electrooculography (EOG) signals, aimed at advancing the field of assistive human–machine interaction. Unlike prior work limited to simple cursor movements or binary commands, our approach uses the temporal and spatial richness of EOG data to support dynamic, multi-command robotic control. The system integrates a robust signal processing pipeline combining noise reduction, informative feature extraction, and multiple learning architectures to classify eye gestures with high accuracy. Publicly available EOG datasets were used for training and evaluation, with model robustness ensured via stratified five-fold cross-validation. Among the tested models, deep learning architectures such as LSTM significantly outperformed traditional methods by effectively capturing sequential dependencies in eye movement signals, achieving an accuracy of 93.2%. This work contributes a scalable, generalizable framework to the literature on noninvasive neurotechnology, demonstrating the feasibility of EOG-driven robotic control as a low-cost, accessible alternative for individuals with severe motor impairments.**

***Index Terms*—Electrooculography (EOG); robotic arm control; assistive technology; temporal signal classification.**

I. Introduction

Individuals affected by the loss of voluntary motor control, such as those with spinal cord injuries, amyotrophic lateral sclerosis (ALS), or cerebral palsy, face substantial challenges in independently interacting with their surroundings. Despite advancements in assistive technologies, many existing systems still require some degree of physical input, limiting accessibility for users with complete paralysis. This work addresses this limitation by investigating the use of EOG signals as a basis for robotic assistive control. By leveraging the eye's bioelectrical activity, the proposed system enables noninvasive, hands-free interaction, offering a promising alternative for reliable assistive operation in cases of severe motor impairment.

EOG is a technique used to monitor eye movements by detecting the electrical signals generated by eye activity [1]. The human eye functions as a dipole system, creating measurable voltage changes on the skin's surface as the eyes move, which can be captured using electrodes. These signals provide full directional information, allowing the system to distinguish among five types of gaze patterns: left-right, up-down, and blinking movements. EOG proves to be an effective control method, particularly for individuals with impaired speech and motor function, due to its reliability in guiding robotic limbs. The core contribution of this research is the development of an adaptive framework for robotic arm control using EOG signals, enabled by machine and deep learning models. Unlike prior work that has largely confined EOG-based systems to cursor navigation or binary command execution, this study extends the frontier toward continuous and dynamic control of robotic manipulators. The proposed system integrates support vector machines (SVM) [2], lightweight convolutional neural network (CNN) [3], and LSTM [4] to classify eye movement patterns in real time. These models collectively enable a more nuanced interpretation of user intent, forming the basis of an intelligent and responsive assistive control mechanism.

The structure of this paper is organized as follows: Section II discusses the related work. Section III describes the dataset used in this study. Section IV presents the proposed methodology, followed by Section V, which presents and discusses the experimental results. Finally, Section VI concludes the paper and outlines potential future work.

II. Related Work

EOG has gained significant attention as a tool for human–computer interaction, particularly in assistive technologies. The eye functions as a dipole, with the cornea exhibiting a more positive charge relative to the retina. This polarity generates voltage changes during eye movement, which can be captured by surface electrodes. Compared to electroencephalography (EEG) and electromyography (EMG), EOG offers simpler calibration, reduced complexity, and more reliable signal acquisition, making it a suitable candidate for real-time assistive systems. In [5], Alex et al. proposed a deep learning-based framework for robotic arm control using brain-computer interface (BCI) technology. In this approach, motor imagery EEG signals were acquired from three targeted electrode sites and subjected to a two-stage denoising process involving a Butterworth bandpass filter and Symlet-9 Wavelet Packet Decomposition. From the preprocessed signals, statistically significant features were extracted to represent neural activity patterns associated with motor intent. These features were then used to train a multi-layer perceptron (MLP) deep neural network, enabling accurate classification of imagined hand movements. The system achieved a cross-validated clas-

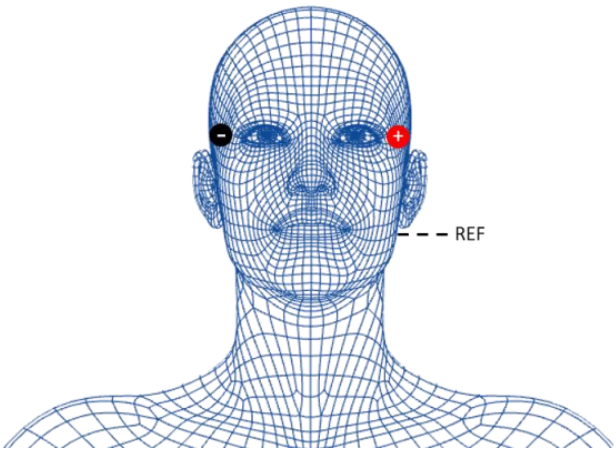
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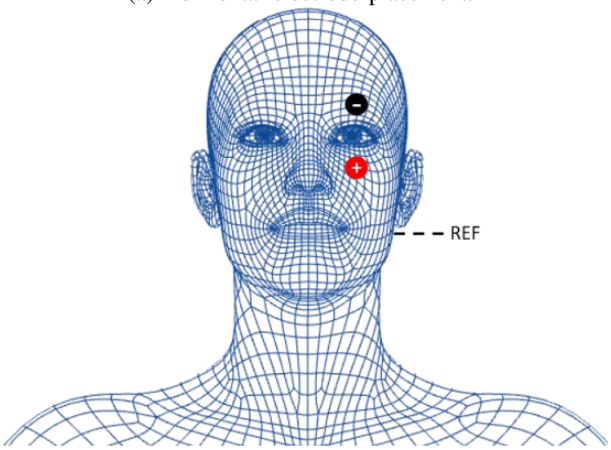
sification accuracy of 85.41% using k-fold validation and demonstrated robust real-time control of a robotic arm through seamless integration with Arduino-based hardware, validating its effectiveness for neuroprosthetic applications. Sekkat et al. [6] proposed a vision-based control framework for robotic arm manipulation that integrates deep reinforcement learning with advanced perception and control modules. The system employs a Deep Deterministic Policy Gradient (DDPG) algorithm to learn optimal grasping strategies, while leveraging YOLOv5 for real-time object detection and inverse kinematics for accurate joint angle computation. Deployed on a five-degree-of-freedom robotic arm, the architecture enables autonomous target localization, motion planning, and grasp execution, demonstrating a cohesive pipeline for end-to-end robotic manipulation in unstructured environments. Satheeshkumar and Arivoli [7] developed a real-time robotic arm control system that measures and replicates human hand grip gestures using flex sensors. Their approach ensures that the grip value of the robotic hand closely matches that of the human hand, enhancing precision in object handling tasks. The system features a five-fingered robotic hand with 14 degrees of freedom, and camera feedback is used to monitor robotic hand activities remotely. This design enables the robotic hand to mimic human grip for various objects, aiming to achieve natural and accurate object manipulation similar to a real human hand. Sharma et al. [8] introduced an alternative method for detecting eye closing and opening using EOG signals in real-time, achieving high detection accuracies for eye closing and opening. When combined with eye blink detection, the system reached an overall accuracy of 96.9%. This EOG-based detection was then applied to control a robotic arm for pick-and-place tasks. The results demonstrated that the EOG interface is both accurate and practical, using simple eye movements such as closing, opening, and blinking. Ayse Nur Ay Gul et al. [9] demonstrated that EOG signal features can be effectively classified for robotic arm control using conventional machine learning models, with KNN achieving notably high accuracy. While their work establishes a strong baseline, it primarily focuses on static classification performance within a limited gesture set. In contrast, our study aims to advance the field by exploring the capacity of temporal and spatial deep learning architectures, namely CNN and LSTM, to model sequential dynamics in EOG signals. Rather than optimizing a single model for maximum accuracy, we conduct a systematic comparison across architectures to evaluate their robustness, generalizability, and suitability for real-time assistive control. Additionally, our framework expands the gesture vocabulary to enable multi-action robotic manipulation, thereby addressing the need for more nuanced and flexible control in practical assistive applications.

III. Data Acquisition

The proposed approach improves generalizability between users and enables real-time translation of EOG signals into robotic arm movements. It uses the publicly available BNCI Horizon 2020 Dataset 004-2014, which includes multichannel EOG recordings from 12 subjects [10]. The dataset consists of approximately 50,000 labeled samples, each containing 256 time steps across four EOG channels. The participants performed eye movements in five directions: left, right, up, down, and blink. The electrodes were placed on the lateral canthi to detect horizontal movements and above and below the eyes for the detection of vertical movements, as shown in Figure 1. Data acquisition was performed at a sampling rate of 256 Hz. The dataset provides high temporal resolution and robust labeling for each class, supporting the development of accurate classification models. The signals were segmented into one-second windows (256 samples per channel), normalized using z scores, and filtered with a bandpass at 0.1-30 Hz. An 80/20 train-test split was applied within each five-fold stratified cross-validation. Segments containing noise from blinks, head movements, or muscle contractions were excluded on the basis of amplitude thresholds.



(a) Horizontal electrode placement.



(b) Vertical electrode placement.

Fig. 1: Illustration of electrode placements for EOG data acquisition.

Figure 2 illustrates typical EOG signal profiles for horizontal (left/right), vertical (up/down), and blink actions. Horizontal gaze patterns manifest as sinusoidal waves, vertical movements exhibit higher-frequency oscillations, and blinks

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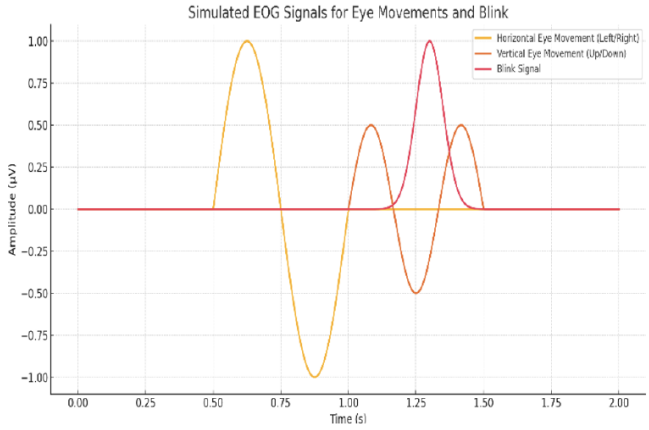


Fig. 2: Simulated EOG signals representing distinct eye movement patterns: horizontal eye movement (left/right) shown in orange, vertical eye movement (up/down) shown in pink, and blink activity shown in red. Each signal is characterized by its unique waveform, amplitude, and timing over a 2-second interval, illustrating the separability of ocular events for classification purposes.

produce sharp, spike-like responses. These distinct waveform characteristics form the foundation for the proposed classification system.

IV. Proposed Methodology

The signal processing pipeline begins with the application of a butterworth bandpass filter to isolate relevant EOG signal components and eliminate noise outside the target frequency range [11]. Following filtering, each segmented window undergoes normalization to mitigate baseline drift and reduce intersubject variability. For traditional classification using SVM, feature extraction was performed in the time domain. Extracted features included mean, standard deviation, signal energy, zero-crossing rate, skewness, and kurtosis, which are commonly used for capturing temporal and statistical characteristics of biosignals. The SVM classifier employed a radial basis function (RBF) kernel, defined as:

$$K(\mathbf{X}_i, \mathbf{X}_j) = \exp\left(-\gamma \|\mathbf{X}_i - \mathbf{X}_j\|^2\right) \tag{1}$$

In this equation, $K(\mathbf{X}_i, \mathbf{X}_j)$ represents the similarity between two feature vectors $\mathbf{X}_i$ and $\mathbf{X}_j$. The term $\|\mathbf{X}_i - \mathbf{X}_j\|^2$ is the squared Euclidean distance between the two vectors. The parameter $\gamma = 0.01$ controls how quickly the similarity decreases with distance. A larger γ value makes the function focus on nearby points, while a smaller γ gives a broader similarity range. The exponential function ensures that the kernel value lies between 0 and 1, where 1 means the vectors are identical. The prediction function is given by

$$y_{svm} = \text{sign}\left(\sum_{i=1}^{N} \alpha_i y_i K(\mathbf{X}_i, \mathbf{X}) + b\right) \tag{2}$$

where y_{svm} is the predicted class label for the input feature vector $\mathbf{X}$. The summation considers all training samples $\mathbf{X}_i$, each weighted by a Lagrange multiplier α_i and class label y_i. The kernel function $K(\mathbf{X}_i, \mathbf{X})$ measures similarity between the input and training data, typically using the RBF kernel. The bias term b adjusts the decision boundary. The sign function determines the final class by checking whether the result is positive or negative.

In contrast, the CNN and LSTM models received raw, segmented signal data directly, avoiding extensive preprocessing. The CNN architecture consisted of three one-dimensional convolutional layers, each followed by ReLU activation functions and max-pooling layers to progressively downsample and extract hierarchical features. Dense layers were added to the model with dropout regularization to prevent overfitting. The LSTM architecture consists of two LSTM layers, each with 64 memory units, allowing the network to effectively learn temporal dependencies in sequential EOG data. The models were trained using the Adam optimizer with a learning rate of 0.001. A softmax activation function was used in the final layer to perform multi-class classification. Finally, each predicted class y_{cnn} and y_{lstm} was assigned to the corresponding control commands C_{cnn} and C_{lstm} for the robotic arm. These commands were transmitted through a serial communication interface, enabling the robotic system to respond to the user's eye movements in real time, as described in Algorithm 1.

The algorithm begins by taking preprocessed EOG signals as input features X. The trained classification models are then loaded, SVM, CNN, and LSTM, each tasked with recognizing and classifying distinct eye movement gestures. The yaw (θ_y) and pitch (θ_p) angles of the robotic arm are initialized to $45°$ and $30°$, respectively. At each time step i, the input is passed to the models to obtain the predictions: $y_{\text{svm}}, y_{\text{cnn}}, and y_{\text{lstm}}$. These predictions are used to simulate movement decisions, with each model controlling the robotic arm separately for evaluation. Based on the command (C_{model}), the corresponding angle is updated using a fixed increment $\Delta = 1.5$. The forward kinematics are then calculated using trigonometric expressions involving link lengths L_1, L_2, and angles θ_y, θ_p, to obtain the 3D coordinates (x_2, y_2, z_2) of the end effector. These coordinates are visualized to animate the 2-DOF arm movement and stored for later analysis.

V. Results and Discussion

This section presents a comparative evaluation of the three machine learning models, SVM, CNN, and LSTM, applied to EOG-based robotic arm control. The analysis emphasizes key performance metrics such as accuracy, precision, recall, and F1 score to assess each model's effectiveness in classifying eye gestures. The results underscore the distinct advantages and limitations of each approach, providing valuable insights into their feasibility for real-time, hands-free assistive technologies.

Based on the results shown in Table I, the LSTM model demonstrated the highest overall performance for EOG-based robotic arm control, achieving an accuracy of 93.2% and an F1 score of 92.8%. This highlights LSTM's effectiveness in capturing temporal dependencies within the EOG signal

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Algorithm 1: EOG based Control of 2-DOF Robotic Arm

Input: Preprocessed EOG features X
Output: 2-DOF Arm Animation, Movement Log
Load trained models: SVM, CNN, LSTM
Initialize: angles $\theta_y \leftarrow 45°$, $\theta_p \leftarrow 30°$
Link lengths: L_1, L_2; Time steps: N
for $i \leftarrow 1$ **to** N **do**
  Predict gesture:
    $y_{\text{svm}} \leftarrow \text{SVM}(X_i)$
    $y_{\text{cnn}} \leftarrow \text{CNN}(X_i)$
    $y_{\text{lstm}} \leftarrow \text{LSTM}(X_{1:i})$
    Predicted command from SVM: $C_{\text{svm}} \leftarrow y_{\text{svm}}$
    Predicted command from CNN: $C_{\text{cnn}} \leftarrow y_{\text{cnn}}$
    Predicted command from LSTM: $C_{\text{lstm}} \leftarrow y_{\text{lstm}}$
    For each model: SVM, CNN, LSTM
      Use predicted command: C_{model}
      Update angles:
      if C_{model} == *Right* **then**
        $\theta_y \leftarrow \theta_y + \Delta$
      else if C_{model} == *Left* **then**
        $\theta_y \leftarrow \theta_y - \Delta$
      else if C_{model} == *Up* **then**
        $\theta_p \leftarrow \theta_p + \Delta$
      else if C_{model} == *Down* **then**
        $\theta_p \leftarrow \theta_p - \Delta$
      Compute Forward Kinematics:
      $x_1 = L_1 \cos(\theta_y) \cos(\theta_p)$
      $y_1 = L_1 \sin(\theta_y) \cos(\theta_p)$
      $z_1 = L_1 \sin(\theta_p)$
      $x_2 = x_1 + L_2 \cos(\theta_y) \cos(\theta_p)$
      $y_2 = y_1 + L_2 \sin(\theta_y) \cos(\theta_p)$
    $z_2 = z_1 + L_2 \sin(\theta_p)$
    Visualize arm and save (x_2, y_2, z_2)

TABLE I: Performance comparison of SVM, CNN, and LSTM models for EOG-based robotic arm control across key classification metrics.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score
SVM	82.0	81.5	80.7	81.1
CNN	90.3	89.9	90.8	90.3
LSTM	93.2	93.0	92.7	92.8

sequences. The CNN model also performed strongly, with a 90.3% accuracy and 90.3% F1 score, showcasing its strength in extracting spatial features from raw input data. While the SVM model had the lowest performance (accuracy of 82.0%), its lightweight structure makes it computationally efficient, especially for systems with limited hardware resources. These findings suggest that LSTM is best suited for applications requiring high classification precision, whereas CNN offers a good trade-off between performance and efficiency. The results reinforce the effectiveness of deep learning methods, especially those capable of capturing temporal dependencies,

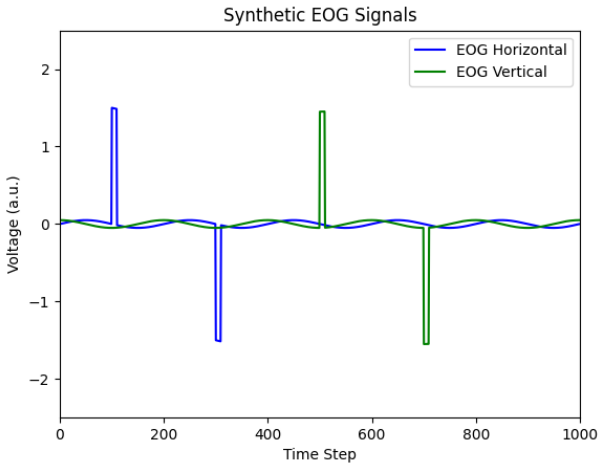


Fig. 3: Simulated EOG signals representing horizontal and vertical eye movement patterns.

such as LSTM networks, in delivering a reliable interpretation of EOG signals for assistive robotic control systems.

To validate the effectiveness of the proposed control system, real-time eye movements simulations were used to drive the robotic arm, as shown in Figure 3. The horizontal and vertical movements were simulated using sine and cosine functions with a frequency of 0.5 Hz and an amplitude of ± 0.05 V. The simulated EOG signals corresponding to the directional eye gestures were fed into the trained machine learning models, and the resulting control commands were visualized through a 2-DOF arm animation. Figure 4 illustrates the movement of the robotic arm end effector in response to four distinct EOG-based control commands: *Up*, *Down*, *Right*, and *Left*. For the *Up* command, the arm exhibits a vertical displacement from approximately $Z = 5$ cm to $Z = 15$ cm, indicating a 10 cm upward motion, while maintaining minimal change in the X and Y directions. The *Down* command results in a reverse movement, with the end effector descending from $Z = 15$ cm to $Z = 5$ cm, reflecting a 10 cm drop. In the case of the *Right* command, the arm shifts horizontally along the X-axis from $X = -10$ cm to $X = 10$ cm, covering a 20 cm rightward path, whereas the *Left* command produces a leftward shift from $X = 10$ cm to $X = -10$ cm. Notably, the Z coordinate remains relatively constant during horizontal motions, and the X and Y coordinates are stable during vertical displacements. These results confirm the system's capability to translate EOG signals into accurate directional robotic arm control, demonstrating its potential for real-time assistive applications.

VI. Conclusion

This study introduced a machine learning framework for EOG-based robotic arm control, designed to provide hands-free interaction for individuals with severe motor impairments. Through a comparative evaluation of SVM, CNN, and LSTM models, the results highlight the superior performance of deep learning approaches, particularly LSTM, in capturing the

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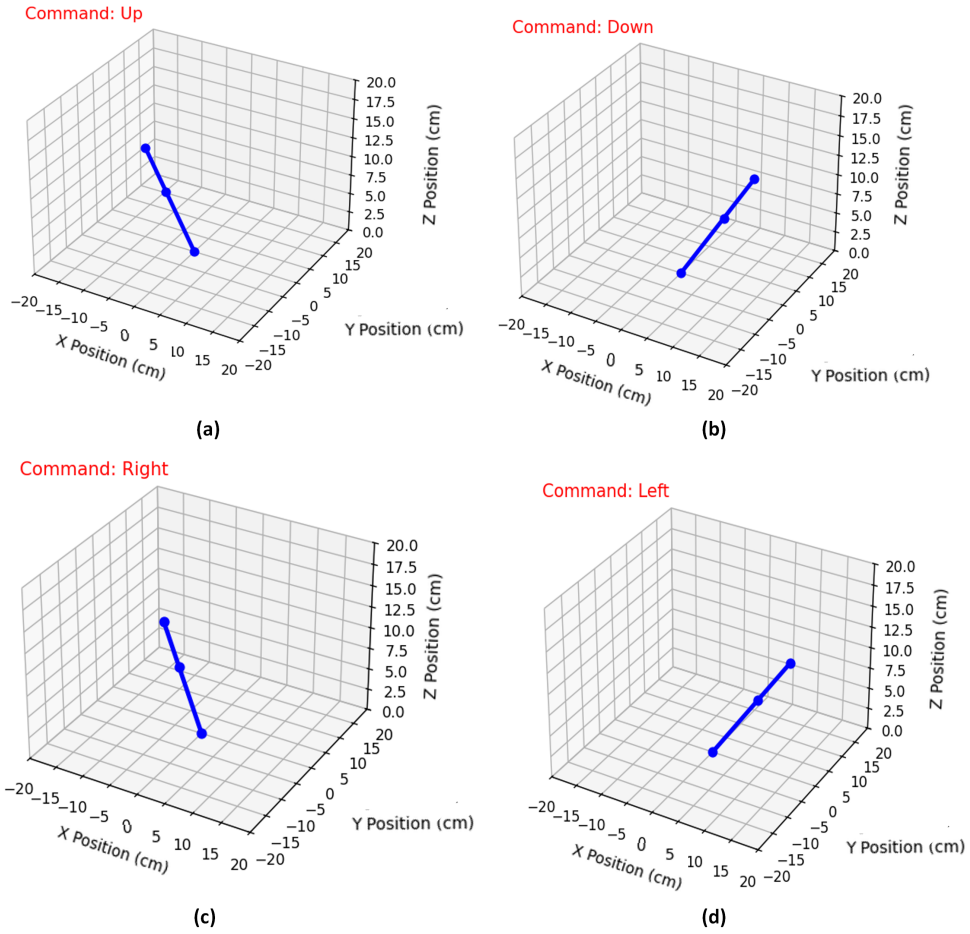


Fig. 4: 3D simulation of robotic arm movements in response to EOG-based commands. (a) Command: Up — the arm moves vertically upward. (b) Command: Down — the arm moves vertically downward. (c) Command: Right — the arm shifts along the positive X-axis. (d) Command: Left — the arm shifts along the negative X-axis. Each trajectory is plotted in 3D space showing the end-effector position in centimeters.

temporal dynamics of EOG signals for accurate and responsive command execution. The system's structured pipeline, integrating signal preprocessing, feature extraction, and classification, effectively translates eye movements into actionable robotic gestures.

Beyond demonstrating model accuracy, this work expands on prior literature by evaluating widely used learning architectures under subject-independent conditions using publicly available datasets, thus enhancing the framework's generalizability and robustness. Additionally, the system supports an extended command set, enabling more complex, multi-action robotic manipulation. These findings highlight the practical potential of EOG-driven control systems in developing scalable and adaptive assistive technologies for real-world deployment.

While the results indicate substantial progress, practical implementation remains a challenge. Key issues such as integration with physical robotic platforms, maintaining system performance in real-world environments, and achieving independence from signal quality variability must be addressed.

Future work will focus on three key directions: enhancing the system's scalability, conducting user-centered evaluations in clinical and home environments, and refining the integration between hardware and software components. A particularly promising avenue involves extending the current EOG classification framework to enable real-time control of robotic manipulators. This will include embedding the trained machine learning models into an interactive control interface, allowing eye gestures to be mapped directly to specific robotic actions, such as arm movement, gripping, or object release, thereby advancing the system's applicability in real-world assistive scenarios. In parallel, the latency of the proposed model will be systematically explored and optimized to ensure seamless real-time responsiveness, while energy efficiency will be analyzed as a critical factor for deploying the system in portable or wearable assistive devices.

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